

An interpretable artificial intelligence framework for defining the operational basin of full-scale membrane bioreactors in semiconductor wastewater treatment

Yujae Jeon^a, Duc Anh Nguyen^b, Kim Anh Nguyen Thi^b, Quoc Thai Nong^b, Am Jang^{b,c,*}

^a Department of Semiconductor Display Engineering, Sungkyunkwan University (SKKU), (16419) 2066, Seobu-ro, Jangan-gu, Suwon, Gyeonggi-do, Republic of Korea

^b Department of Global Smart City, Sungkyunkwan University (SKKU), (16419) 2066, Seobu-ro, Jangan-gu, Suwon, Gyeonggi-do, Republic of Korea

^c SKKU Institute of Energy Science and Technology (SIEST), Sungkyunkwan University (SKKU), (16419) 2066, Seobu-ro, Jangan-gu, Suwon, Gyeonggi-do, Republic of Korea

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ABSTRACT

Full-scale membrane bioreactors treating semiconductor wastewater operate under conditions laboratory studies rarely reproduce: fluctuating organic and ionic loads tied to production schedules, specialty solvents and chelating agents from photoresist stripping and CMP, and extended sludge ages far outside the short-SRT regimes on which most fouling theory is based. Operators must simultaneously control transmembrane pressure (TMP), permeate throughput, and hydraulic balance, yet no data-driven framework has identified which operating combinations reliably achieve this. We developed an interpretable machine-learning framework using 4593 hourly SCADA records from a 1125 m³/h industrial MBR to define its operational basin. Sixteen algorithms across six families were benchmarked for TMP, permeate flow, and membrane-tank level. Extra Trees achieved the highest accuracy for all targets ($R^2 = 0.988, 0.933, 0.908$) and retained first rank under blocked, chronological, and cross-train validation, including an independent parallel train. When refit daily, it held $R^2 = 0.871$ for TMP over a four-month out-of-time period. SHAP analysis identified sludge retention time as the dominant driver, with fouling worsening monotonically across the plant's extended-aeration range. Feasibility mapping using local-covariance resampling constrained to the plant's attainable operating manifold identified 207,238 feasible states out of 500,000 sampled. Model-derived feasibility tracked measured attainment across 70 bins ($r = 0.99$), and operation inside recommended bands met all three targets 61–82% of the time versus 37% baseline. A window derived from the first seven months raised target attainment to 61.9% in the final three months (odds ratio 3.89). The framework yields an actionable control protocol, SRT 43.2–72.2 days, HRT 5.9–6.4 h, MLSS 5580–6138 mg/L, F/M 0.023–0.025 day⁻¹, aeration 4868–5892 m³/h, and is transferable to other industrial MBRs with site-specific retraining.

1. Introduction

The semiconductor manufacturing industry is one of the most water-intensive sectors globally [1–6], consuming between 1400 and 4200 L of ultrapure water per square centimeter of fabricated wafer [7–10]. As device geometries shrink and production volumes rise, securing supply while managing contaminated effluent has become critical, and the effluent itself demands multi-barrier treatment [11–13]: organic compounds from photoresist processes, ionic contaminants from etching and

cleaning, and colloidal particles from polishing must all be removed, with each unit process operating reliably to safeguard the ones downstream [9,14]. Membrane bioreactors (MBRs) have become the preferred solution because they combine biological degradation with high-efficiency membrane separation in one compact process. In a submerged configuration, ultrafiltration (UF) membranes sit directly in the activated sludge bioreactor in place of a secondary clarifier, producing permeate largely free of suspended solids, bacteria and high-molecular-weight organics. The configuration gives effluent quality

* Corresponding author at: Department of Global Smart City, Sungkyunkwan University (SKKU), (16419) 2066, Seobu-ro, Jangan-gu, Suwon, Gyeonggi-do, Republic of Korea.

E-mail addresses: junyujae@naver.com (Y. Jeon), nguyenducanh@g.skku.edu (D.A. Nguyen), toikimanh@gmail.com (K.A.N. Thi), thainq@g.skku.edu (Q.T. Nong), amjang@skku.edu (A. Jang).

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independent of sludge settleability, a smaller footprint, operation at elevated mixed liquor suspended solids (MLSS), and decoupling of solids retention time (SRT) from hydraulic retention time (HRT), which enriches slow-growing nitrifiers and enhances nutrient removal [15–19].

In semiconductor wastewater reclamation, MBRs act as the pretreatment barrier upstream of reverse osmosis (RO) polishing [20,21], and the quality of the UF permeate sets the fouling trajectory of the downstream RO membranes [22,23]. Fouling remains the dominant operational constraint, and it is more complex in submerged MBRs than in pressure-driven standalone UF because the membrane sits in mixed liquor carrying biomass, extracellular polymeric substances (EPS), soluble microbial products (SMP) and colloidal organic matter [24–26]. Its rate is set by an interacting group of factors: MLSS and sludge rheology govern cake formation, the food-to-microorganism (F/M) ratio and SRT control EPS and SMP production, aeration supplies the shear that mitigates cake build-up, and HRT modulates organic loading [27–29]. Semiconductor wastewater imposes a fouling signature unlike municipal streams in three respects. The organic load is dominated not by readily biodegradable substrate but by photoresist strippers, developers, solvents and chelating agents, a recalcitrant fraction of which adsorbs preferentially onto PVDF to form a poorly reversible layer [30–34]. Chemical-mechanical polishing contributes silica and sub-micron abrasive colloids that seed a dense, low-porosity cake [35–37]. And loading follows the fab's batch production calendar rather than diurnal human activity, so organic and nitrogen pulses arrive on a schedule that is steep and only weakly predictable from the wastewater side. Because carbon is often limited for denitrification in these nitrogen-rich streams, glucose dosing becomes a routine control lever that couples the carbon-to-nitrogen balance directly to fouling [38,39]. Together these features place the plant in an extended-aeration, high-SRT regime in which conventional short-SRT heuristics do not transfer, and they create a non-linear process landscape that defies simple correlation and rule-based optimisation.

Ubiquitous SCADA sensing and advances in machine learning (ML) have opened data-driven prediction and optimisation of MBR operation [40–42]. Artificial neural networks have predicted TMP from MLSS, flux and aeration rate [43]; random forest and gradient boosting have predicted fouling in pilot-scale MBRs with R^2 above 0.90 [44]; long short-term memory networks have captured the temporal dynamics of TMP [45]; support vector regression and random forest have been applied to permeate quality [46]; and XGBoost has related operating conditions to fouling rates in municipal MBRs with reported R^2 of 0.94–0.97 [47]. Three limitations hinder translation into industrial practice. Most studies use laboratory- or pilot-scale systems fed synthetic or municipal wastewater [48,49], which cannot reproduce full-scale complexity, where feedwater composition follows manufacturing schedules, cleaning residuals perturb the biology and equipment degradation introduces non-stationarity [50]. Most treat the model as a black box, reporting accuracy while offering little insight into which parameters govern membrane performance [51,52], and without that transparency a model cannot inform decisions or earn the trust of the operators who must act on it. And, less often acknowledged, hourly plant records are strongly autocorrelated, so the random train-test partitions used almost universally in this literature place near-duplicate observations on both sides of the split, making reported accuracies difficult to read as performance on genuinely unseen operating conditions.

Two gaps follow. The first concerns interpretability. Explainable artificial intelligence methods, principally SHAP and partial dependence plots, have been applied across wastewater treatment, including aeration control, nutrient removal and anaerobic digestion monitoring, but their application to MBR fouling is limited and confined to single models and narrow parameter ranges. What is missing is multi-model interpretability, in which SHAP is applied across algorithmic families to expose model-dependent bias in the importance rankings [53]; such bias matters because different architectures can assign conflicting importance to the same variable [54], so the recommendation depends on

which model is consulted, a phenomenon documented for RO systems but not for MBRs. The second gap concerns the step from prediction to guidance. Existing ML-based optimisation of membrane processes either pairs a trained model with a search algorithm to locate point optima [55], or derives qualitative guidance from feature importance without formal optimisation [56]. The first returns single solutions without indicating how robust they are or how wide the feasible envelope is; the second gives understanding without quantitative bounds. Neither approach, moreover, takes account of the fact that the operating variables of a full-scale plant are strongly interdependent, so that a feasible region is informative only if it is mapped over the joint operating space the facility can actually reach. For semiconductor MBR systems, where biology and filtration are coupled through shared variables such as MLSS, SRT and aeration, a unified framework is needed that treats TMP, permeate flow and membrane tank level simultaneously and that samples only states the plant can actually reach.

This study introduces an interpretable artificial intelligence framework for optimising MBR operation in semiconductor wastewater treatment, developed from 4593 hourly measurements collected over 306 consecutive days at a full-scale industrial facility in Asan City, Republic of Korea. The MBR, comprising submerged PVDF ultrafiltration membranes integrated with a multi-stage biological process, is the pretreatment step ahead of RO polishing in a reclamation train rated at 1125 m³/h. Unlike laboratory datasets acquired under controlled conditions, the analysis uses SCADA records that carry the full complexity of industrial operation, including variable organic loading from upstream manufacturing, routine maintenance and recovery cleaning events, and progressive membrane ageing over the record.

The complete framework is set out stage by stage in Fig. 1, and the work makes three contributions. First, it provides the most comprehensive ML benchmarking reported for semiconductor-wastewater MBRs, sixteen algorithms across six families predicting TMP, permeate flow and membrane tank level simultaneously, evaluated under four validation designs so that genuine predictive skill is separated from the accuracy conferred by temporal autocorrelation. Second, it applies SHAP across all sixteen models and distinguishes consensus process conclusions from artefacts of a particular learner, a distinction unavailable from the single-model analyses that dominate this literature. Third, it maps the feasible operating region by manifold-constrained resampling rather than independent sampling of marginal ranges, and validates the resulting window against the plant's measured attainment rate and out of time on months the models never saw. The framework is directly transferable in method, though quantitative windows are site-specific and require retraining on facility data.

2. Models and methods

2.1. System description and data collection

The experimental data were collected from a full-scale submerged membrane bioreactor (MBR) system at a semiconductor wastewater treatment plant in Asan City, Republic of Korea. The facility utilizes an anoxic-oxic-anoxic-oxic (A-O-A-O) biological configuration, followed by submerged UF membranes, which combine biological nutrient removal with solid-liquid separation in a single integrated process. The MBR system operates at a design capacity of 1125 m³/h, treating semiconductor fabrication wastewater before it is sent to the downstream reverse osmosis system. The UF system consists of 9700 polyvinylidene fluoride (PVDF) hollow-fibre membrane modules manufactured by SUEZ, providing a total membrane area of 329,800 m². Detailed system specifications and sensor information are available in Section S1 and Table S1 of the supplementary document.

The membrane tank and biological reactor volumes are reported here for the first time so that the hydraulic terms can be reproduced. The A-O-A-O bioreactor has working volumes of 3841 m³ for the anoxic zones and 8604 m³ for the oxic zones, together with a 376 m³ pH

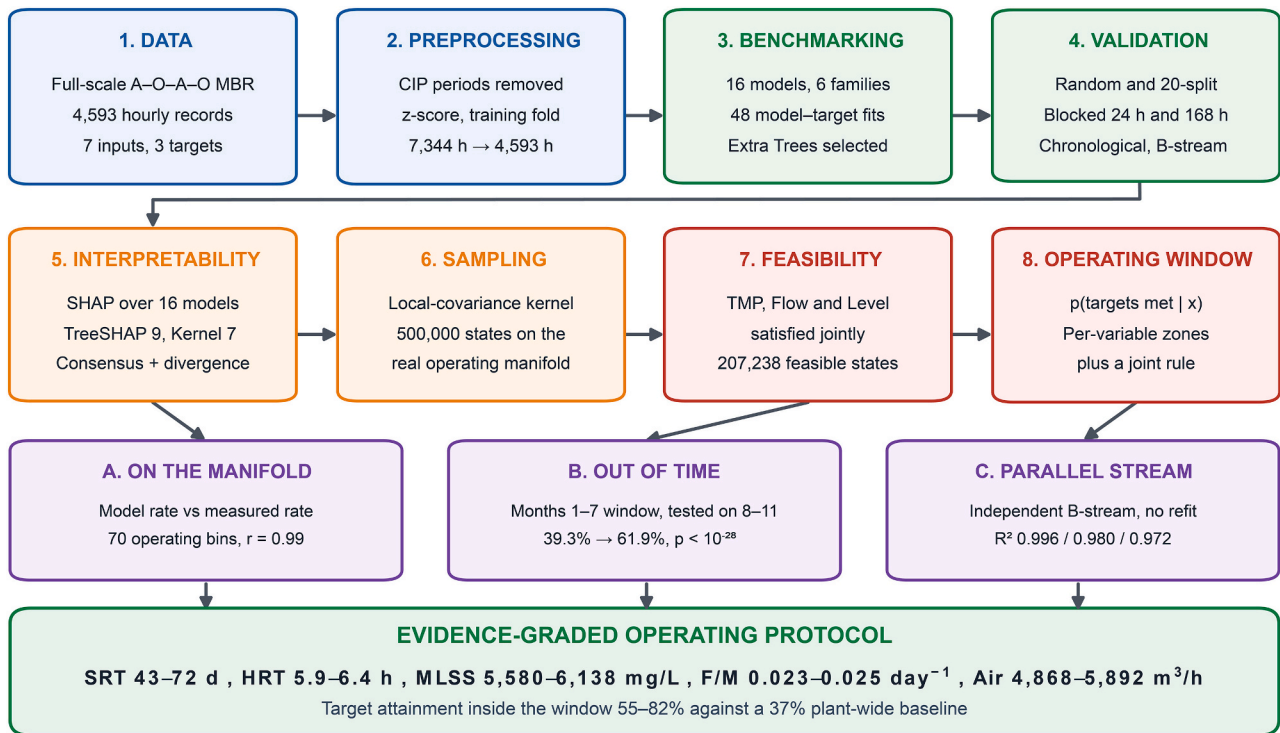


Fig. 1. Workflow of the interpretable machine-learning framework, from SCADA acquisition through model benchmarking and multi-design validation, cross-model SHAP interpretability, manifold-constrained feasibility sampling and conditional-rate window derivation, to the three independent validations and the resulting operating protocol. Follow the numbered stages left to right and top to bottom; the three purple boxes are independent checks on the result, and the green bar is the operating protocol they support. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

adjustment tank and a 1437 m³ membrane tank. The HRT used in this study refers to the hydraulic retention time of the oxic tanks and is recorded as a virtual parameter (VP) automatically calculated by the plant control system. The mean HRT over the study period was 7.33 h.

Data was collected continuously from 1 January 2025 to 3 November 2025, spanning 306 consecutive operational days. Process parameters were logged at five-minute intervals by the plant's supervisory control and data acquisition (SCADA) system and aggregated to hourly averages, giving 4593 hourly observations after exclusion of chemical cleaning-in-place (CIP) periods, comprising maintenance cleaning (MC) and recovery cleaning (RC) events, which do not represent normal fouling behaviour. Because the record covers ten consecutive months rather than a full annual cycle, no seasonal decomposition was attempted and no seasonal effect is claimed anywhere in this work; the long-timescale variation present in the record is treated as progressive operational drift, principally membrane ageing and the evolving cleaning history, rather than as a seasonal signal.

2.2. Process parameters and operating ranges

Seven input parameters describing biological treatment conditions and membrane operating state were monitored continuously: glucose dosing rate (Glu, a surrogate for supplemental organic loading), mixed liquor suspended solids (MLSS), aeration rate (Air), food-to-microorganism ratio (F/M), the ratio of total organic carbon to total nitrogen (C/N), hydraulic retention time (HRT) and sludge retention time (SRT). Three target variables were predicted: transmembrane pressure (TMP) as the primary fouling indicator, permeate flow rate (Flow) as the measure of productive output, and membrane tank water level (Level) as an indicator of hydraulic balance. The abbreviations defined in this paragraph, namely Glu, MLSS, Air, F/M, C/N, HRT, SRT, TMP, Flow and Level, are used consistently thereafter in the text, tables and figures, including the abstract and keywords. The carbon-

to-nitrogen ratio is measured as TOC/TN; the two terms denote the same quantity and only C/N is used hereafter. Table 1 summarises the statistical characteristics of all measured parameters.

2.3. Data preprocessing and standardisation

Following quality control, input features and output targets were standardised using z-score normalization to ensure that all variables contributed equally to model training, regardless of their original measurement scales. The visualization and distribution of features before and after normalization are shown in Fig. 2.

Standardisation parameters were computed exclusively from the training partition and applied to the test set to prevent information leakage; the formulation is given in Supplementary Section S2. The step also clarifies the structure of the record. The time series of Flow, TMP and Level show substantial short-term variability together with slower drift, so the system exhibits both gradual operational movement and abrupt transitions across all three outputs (Fig. 2A). Standardisation preserves the shape of each empirical distribution, including skewness, multimodality and tail behaviour (Fig. 2B), while placing variables

Table 1

The statistical characteristics of all measured parameters.

Feature	Count	Mean	Std	Min	Max
Glu (L/min)	4593	0.6775	0.1807	0.4027	1.3515
MLSS (mg/L)	4593	4889.5	1075.2	1919.4	7628.8
Air (m ³ /h)	4593	5945.7	534.5	4469.1	7268.3
F/M (1/day)	4593	0.0275	0.0078	0.0118	0.0658
C/N ratio	4593	9.4314	2.2581	4.8040	17.5029
HRT (h)	4593	7.3346	0.7947	5.8738	11.0411
SRT (day)	4593	87.828	14.564	43.323	108.333
TMP (bar)	4593	-0.122	0.090	-0.462	-0.034
Flow (m ³ /min)	4593	1.817	0.257	0.618	2.289
Level (lv) (%)	4593	65.774	1.114	64.005	73.942

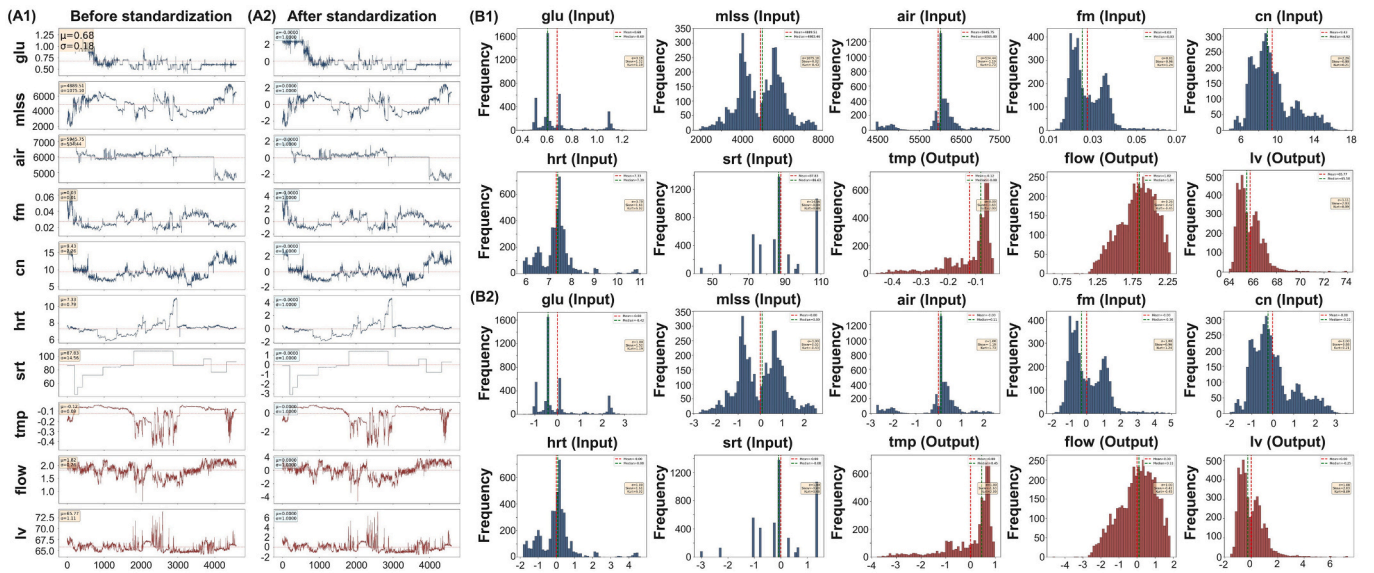


Fig. 2. (A) Time series and (B) distribution of the input and output features of the MBR mainstream (1) before and (2) after normalization. Input features are in blue; output features are in red. The upper row is the raw time series and the lower row the same variables after standardisation; comparing them shows that scaling changes magnitude but not shape. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

whose raw ranges and units differ substantially on a common zero-mean, unit-variance scale so that their magnitudes and variability become directly comparable.

The Pearson correlation matrices before and after normalization confirm that z-scoring leaves the linear relationships among variables unchanged (Fig. 3), preserving dependencies such as the negative association between TMP and Flow, the coupling between Level and loading-related variables, and the hydraulic relationships involving HRT and SRT that jointly constrain Flow and Level. The correlations involving Level emphasise its role as an integrative state variable reflecting both hydraulic conditions and biomass accumulation rather than a purely independent target.

2.4. Data split and robustness evaluation

Because a continuously logged hourly record carries temporal structure by construction, validation was deliberately made more demanding than a single partition, and four complementary schemes were applied. A random 70/30 partition (3215 training and 1378 test samples) is the design used throughout this literature and is retained for comparability with it. A multi-split analysis over 20 independent random partitions, retraining all 16 models each time, quantifies the variability attributable to partitioning. Blocked partitions divide the record into contiguous blocks of 1, 6, 24, 72, 168, 336 and 720 h and assign whole blocks at random, preserving coverage of the operating envelope while separating adjacent hours, so that generalisation from one operating day to another is measured directly. And a strictly chronological partition trains on the first 70% and tests on the final 30%,

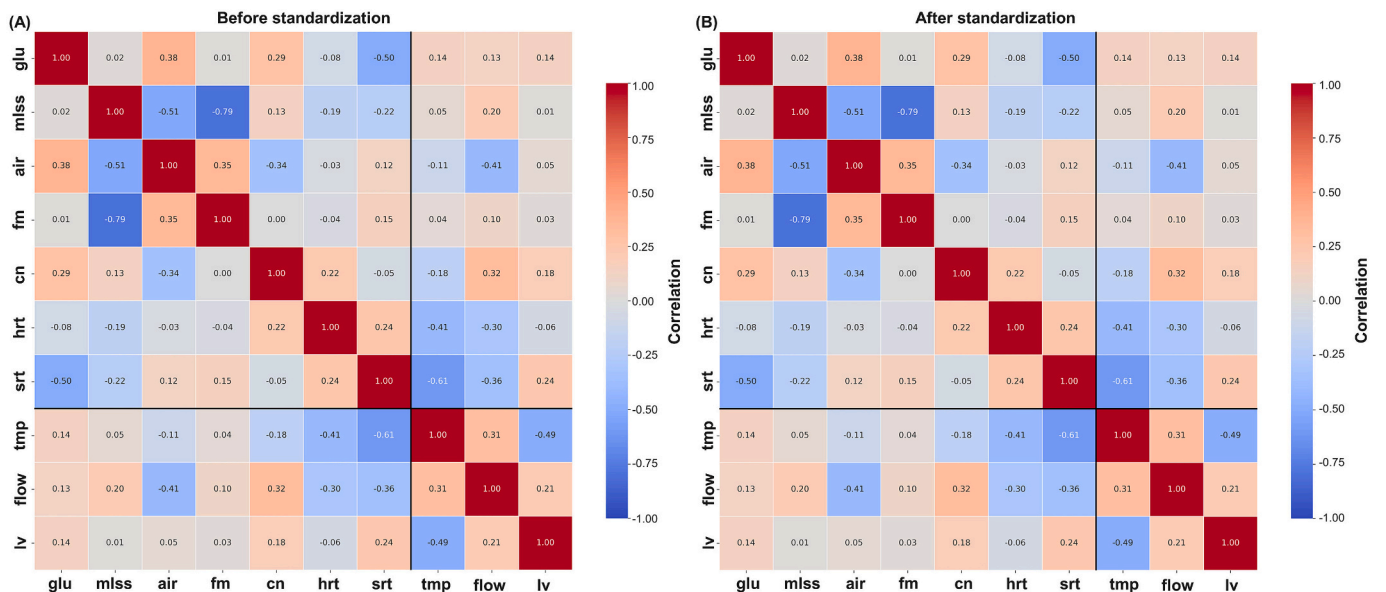


Fig. 3. Relationship of the input and output features of the MBR mainstream (A) before and (B) after normalization. Each cell is a pairwise Pearson correlation, so darker cells mark the variable pairs that move together; comparing the two panels shows that standardisation leaves those relationships unchanged.

with a purged variant discarding a one-week gap at the boundary. An independent set of 4593 hourly observations from a parallel B stream within the same facility assesses transfer to a physically distinct membrane system without retraining. Standardisation statistics were computed on the training fold only in every design. The dependence structure of the record and the full results under each design are tabulated in Supplementary Tables S9 to S12.

2.5. Machine learning model selection and training

Sixteen regression algorithms spanning six modelling families were evaluated for the three targets from the seven operational parameters (Table 2). The portfolio was not assembled for its own sake: each family tests a falsifiable hypothesis about the structure of the biology–membrane relationship, so the comparison is a structured hypothesis test rather than an undirected modelling exercise. Linear and regularised linear models test whether the relationship is adequately additive and monotonic, with regularisation additionally probing whether apparent importance is an artefact of collinearity. Support vector regression with an RBF kernel tests whether a smooth global non-linear surface suffices. K-nearest neighbours tests whether fouling behaviour is locally regular in operating space, that is whether similar conditions reliably give similar responses, which is the assumption underlying any operating-window recommendation. Tree-based ensembles test whether threshold effects and feature interactions dominate, as classical fouling theory predicts, and the bagging-versus-boosting contrast within that family tests whether variance reduction or sequential bias reduction is the better response to noisy industrial sensor data. A multi-layer perceptron with two hidden layers tests whether hierarchical representations add anything beyond these. The pattern of which families succeed and which fail is therefore itself the evidence for the structure of fouling in this system, and it is reported in Section 3.1. A single model was carried forward for interpretation and optimisation to keep the framework consistent across the three targets. Formulations, hyperparameters and training configurations for all 16 algorithms are given in Section S3 and Table S2; all work was performed in Python 3.10 using NumPy, pandas, scikit-learn, XGBoost, LightGBM and SHAP.

2.6. Interpretability analysis through multi-model SHAP

SHapley Additive exPlanations (SHAP) analysis was applied across all 16 models and 3 targets, giving 48 interpretability profiles. This multi-model framework reveals whether different algorithms agree on feature importance and exposes model-dependent bias that single-model analyses cannot detect. TreeSHAP was used for the nine tree-based models and KernelSHAP for the seven non-tree models, and beeswarm plots, mean absolute SHAP values and dependence plots were generated for each model–target combination. Because SHAP quantifies each feature's contribution to the model output, it reflects association within the training distribution rather than verified physical causation; the mechanistic explanations offered in Section 3.2 are physically plausible and literature-consistent interpretations, not demonstrated mechanisms.

Table 2
Summary of 16 machine learning algorithms across six model families.

Model family	Algorithms	Count
Linear	Linear Regression	1
Regularised Linear	Ridge, Lasso, Elastic Net	3
Support Vector	SVR (RBF kernel)	1
Machine		
Instance-based	K-Nearest Neighbours	1
Ensemble Tree	Decision Tree, Random Forest, Extra Trees, Bagging, AdaBoost, Gradient Boosting, Hist Gradient Boosting, XGBoost, LightGBM	9
Neural Network	Multi-Layer Perceptron	1

2.7. Optimal operating condition identification

A multi-objective feasibility framework was used to identify operating regions satisfying the target criteria for all three outputs simultaneously. Because the seven operating variables are interdependent in practice, candidates were generated so as to respect that joint structure; the rationale and a comparison against independent sampling of the marginal ranges are given in Supplementary Section S5. Feasibility was mapped by local-covariance kernel resampling constrained to the plant's real joint operating manifold: a seed hour is drawn uniformly from the 4593 logged observations, the local covariance of its 30 nearest neighbours in standardised input space is formed, the seed is perturbed by a Gaussian increment scaled by a bandwidth factor of 0.6 applied to that covariance, and the candidate is rejected if it leaves the observed marginal range of any input or if its nearest-neighbour distance to the real record exceeds the 99th percentile of real-to-real nearest-neighbour distance (0.539 in standardised units). The resulting 500,000 candidates have a median nearest-neighbour distance of 0.121 against 0.119 between real hours, so the sample explores continuously around the manifold without leaving it. A candidate was classed feasible only where all three predictions fell within their target ranges, with predictive uncertainty taken from the dispersion of the 100 individual trees. Windows were then defined from the conditional feasibility rate, the probability that all three targets are met given the value of an input, rather than from the density of feasible points, which reflects how often the plant visited a region and not how well it performed there. Each input was divided into deciles and the recommended band taken as the widest contiguous run whose measured attainment remains within 5 percentage points of the maximum, subject to a minimum support of 250 logged hours; four of the seven bands are unchanged by this criterion, indicating genuinely sharp optima. Kernel density estimation (KDE), a non-parametric method that reconstructs a continuous density by summing a smooth kernel centred on each observation and so avoids the bin-placement dependence of histograms, was used with a Gaussian kernel and Scott's-rule bandwidth. Detailed algorithms are given in Supplementary Section S5.

2.8. Evaluation metrics

Prediction accuracy was quantified using three complementary metrics: the coefficient of determination (R^2), root mean squared error (RMSE) and mean absolute error (MAE). Their mathematical formulations are provided in Supplementary Section S4.

3. Results and discussion

3.1. ML model prediction performance

3.1.1. Output prediction performance

The predictive performance of all 16 models was evaluated for TMP, permeate flow and membrane tank water level. Fig. 4 shows the R^2 heatmap and predicted-versus-observed scatter across model–target combinations, revealing a clear hierarchy in both model capability and target predictability: TMP was predicted most accurately, followed by permeate flow and then water level. Effluent quality was deliberately excluded as a target. Permeate TOC is measured by an online analyser whose 0.03–1000 ppb range leaves it near its detection limit for most of the record, because the UF membrane retains essentially all suspended and high-molecular-weight material, so the signal is dominated by analyser noise. More fundamentally, UF permeate quality in a submerged MBR is governed by membrane integrity, a step-change phenomenon associated with fibre breakage rather than a continuous function of the variables considered here, making it an integrity-monitoring problem rather than a regression target. The three targets selected instead are those the operator can trade against one another hourly.

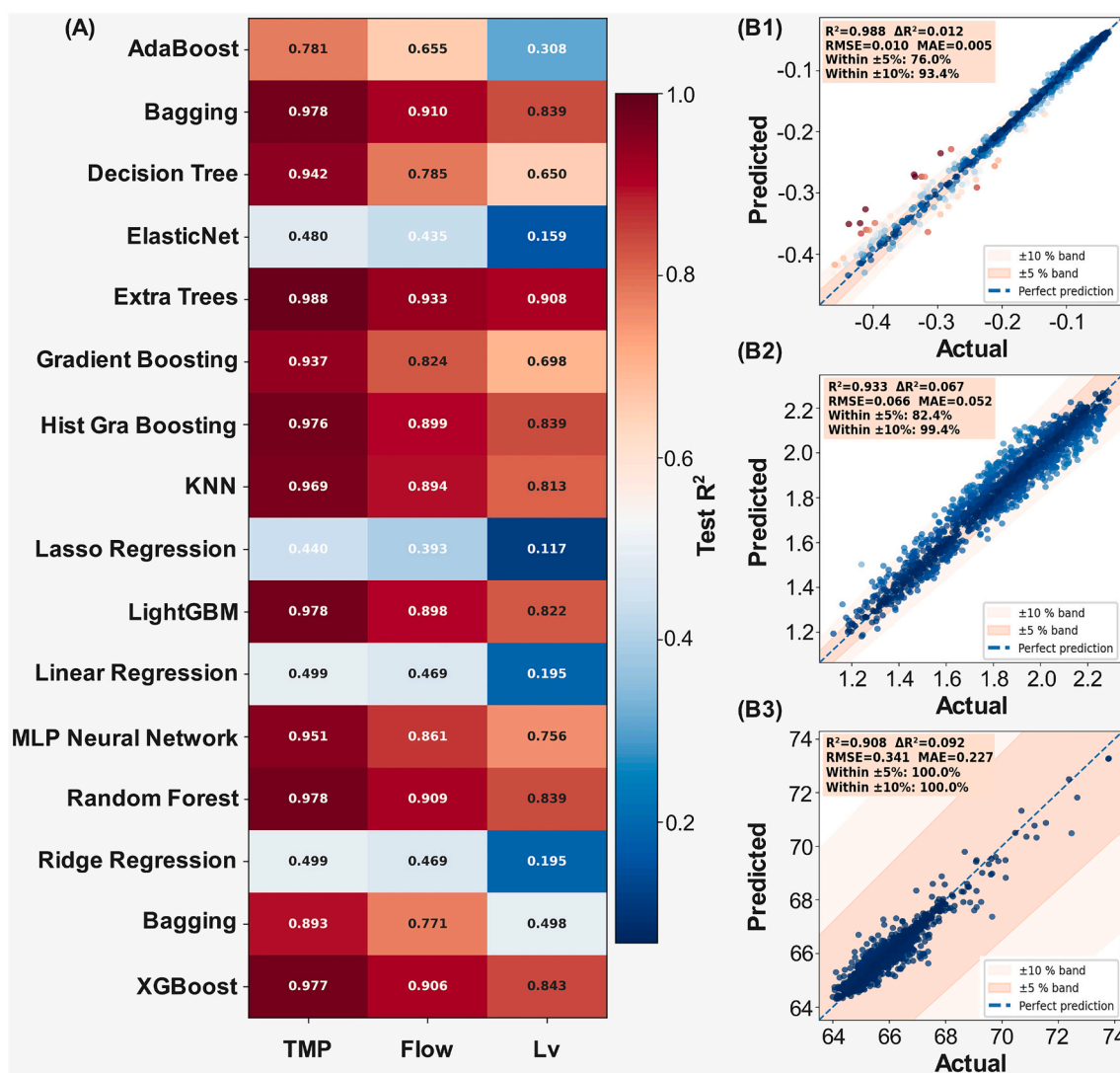


Fig. 4. (A) Test R^2 heatmap of all 16 models and predicted vs. actual scatter plot of the Extra Tree model across three target variables: (B1) TMP, (B2) permeate flow, and (B3) membrane tank water level. Blue indicates low R^2 values, and red indicates high R^2 values in the heatmap. Scatter colour indicates residual magnitude; the dashed line represents perfect prediction; shaded bands indicate $\pm 5\%$ and $\pm 10\%$ error tolerance. In the heatmap each cell is one model-target combination and redder is more accurate; in the scatter panels points should fall on the dashed 1:1 line, and the shaded bands are the 5% and 10% error tolerances. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

For TMP, tree-based ensembles were clearly superior. Extra Trees achieved the highest accuracy ($R^2 = 0.988$, $RMSE = 0.010$ bar, $MAE = 0.005$ bar), followed by Bagging, Random Forest and LightGBM (0.978 each) and XGBoost (0.977) (Fig. 4, Fig. S1), all five above $R^2 = 0.97$. The strong predictability of TMP is physically interpretable: in submerged systems TMP is governed by fouling resistance, which develops progressively with MLSS, aeration scouring intensity and SRT-dependent sludge characteristics, and all of these were supplied to the models directly [28,29,35,36]. TMP is also measured as suction pressure on the permeate side, giving a more stable signal and a higher signal-to-noise ratio than pressurised configurations [35].

Permeate flow prediction followed a similar model ranking but with moderately lower R^2 values. Extra Trees again led ($R^2 = 0.933$, $RMSE = 0.066$ m³/min, $MAE = 0.052$ m³/min), with Bagging ($R^2 = 0.910$), Random Forest ($R^2 = 0.909$), XGBoost ($R^2 = 0.906$), and Hist Gradient Boosting ($R^2 = 0.899$) completing the top five (Fig. 4 and Fig. S2). The reduced accuracy compared to TMP is consistent with the mechanics of MBR operation. Permeate flow is influenced not only by the state of membrane fouling but also by operator-controlled setpoints, backwash cycling, and transient hydraulic disturbances that may not be fully

captured by the hourly-averaged input parameters [23–26]. The influence of intermittent aeration scouring on instantaneous permeate flow introduces variability that high-fidelity models cannot fully account for based solely on the available biological and operational features.

Membrane tank water level proved the most challenging target, the best model reaching $R^2 = 0.908$ (Extra Trees, $RMSE$ 0.341%, MAE 0.227% under the random partition), with the leading ensembles between 0.82 and 0.91 and all other families substantially below. The ordering across the three targets has a direct physical reading. TMP is the most predictable because it integrates fouling resistance over hours to days, and the variables governing that accumulation, principally SRT and MLSS, are slowly varying and were measured directly. Permeate flow is intermediate because it is only partly a membrane property, being set also by throughput demand, backwash cycling and short hydraulic transients that hourly averaging cannot resolve [57–59]. Water level is least predictable because it is a fast hydraulic state governed by the instantaneous balance of influent, permeate extraction, internal recirculation, sludge wasting and level-controlled pump logic, several of which act over minutes and none of which is a model input. The gradient therefore ranks the targets by how far their dynamics are captured by

slow biological state rather than fast control action, implying that model-based planning is most reliable for TMP while water level is better managed by the fast levers identified in Section 3.3. Level accuracy remains operationally useful even so, with MAE below 0.23% against a 2% control window.

3.1.2. Model comparison, consistency and independent validation

A large and systematic gap separated linear from non-linear models across all three targets (Table 3). Linear and Ridge Regression reached R^2 of only 0.499 for TMP, 0.469 for flow and 0.195 for water level, the regularised variants slightly worse as aggressive shrinkage removed small but informative coefficients, while the best ensembles exceeded 0.90 on all three. This is the structured hypothesis test anticipated in Section 2.5: the failure of the additive family together with the success of recursive partitioning is evidence that the biology–membrane relationship is governed by thresholds and interactions rather than proportional effects, consistent with fouling theory in which cake resistance accelerates beyond critical MLSS, aeration shows diminishing returns above a critical scouring velocity, and F/M and SRT interact non-additively [25,29,36,38]. Within the ensemble family the bagging variants consistently outperformed sequential boosting, although XGBoost (0.977) and LightGBM (0.978) nearly matched them for TMP. AdaBoost was markedly worse on every target (0.781, 0.655, 0.308), consistent with the sensitivity of adaptive reweighting to the sensor artefacts of industrial SCADA data, and the advantage of Extra Trees over Random Forest is attributable to the additional randomisation of split thresholds. Among non-tree non-linear models KNN performed strongly (0.969, 0.894, 0.813), indicating that membrane response is locally regular in operating space, the property that makes an operating-window recommendation meaningful at all. Only the Decision Tree for water level showed substantial overfitting (0.890 against 0.650), and the 20-split analysis (Table S3, Figs. S4–S6) confirmed Extra Trees as the best model on mean R^2 and the lowest RMSE and MAE for all three targets.

Transfer to a physically distinct membrane system was assessed on an independent set of 4593 hourly observations from a parallel B stream in the same facility, operating under different influent conditions and with an independent fouling history (Fig. S7). Extra Trees, applied without retraining, achieved $R^2 = 0.996, 0.980$ and 0.972 for TMP, flow and water level (Table S4, Fig. 5A–C), with near-zero mean errors and error standard deviations of 0.014 bar, 0.052 m^3/min and 0.306% (Fig. 5D). The TMP error distribution is slightly right-skewed (skewness 1.025), indicating occasional underestimation under extreme fouling, which is the conservative direction operationally because it prompts earlier rather than later cleaning. The B stream runs over the same calendar period and shares the same influent, so this establishes transfer

across membrane trains rather than across time; temporal transfer is addressed next.

Taken together, the primary test-set performance, the 20-split robustness analysis and the parallel-stream validation identify Extra Trees as the best model for all three targets, and it was carried forward for the interpretability analysis in Section 3.2 and the optimisation in Section 3.3. Using one model for all three targets keeps the operational framework and the feature-importance interpretation consistent across the three performance indicators.

3.1.3. Robustness of the model selection to the validation design

Model selection for a full-scale plant record should not rest on a single partition, so the choice of Extra Trees was tested under four independent designs: the conventional random 70/30 partition, a blocked design in which contiguous blocks are allocated as whole units so that coverage of the operating envelope is preserved while adjacent hours are separated, a strictly chronological partition, and the independent parallel B stream. The outcome is a strong one. Extra Trees ranks first in eight of the nine model–design–target combinations and first for every target under both the random and the 24-h blocked designs (Table S10), and accuracy under 24-h blocking remains high for a full-scale industrial system, with $R^2 = 0.830$ for TMP corresponding to an RMSE of 0.036 bar against an operating range of 0.43 bar and a mean absolute error of 16.3% of the interquartile range of the measured series. Because the record is a continuously logged hourly series and therefore autocorrelated by construction, blocked and random values are reported side by side throughout; the dependence structure, the full per-model results and the divergence between R^2 and absolute error as blocks lengthen are given in Supplementary Section S6 and Fig. S17. More important than absolute accuracy is the stability of the conclusions: SRT remains the first-ranked driver of TMP and of water level under random, blocked 24-h, blocked 168-h and chronological training alike, so the process conclusions are properties of the plant record rather than of any particular partition.

The chronological partition deserves separate comment, because as usually implemented it describes a deployment scenario no plant would adopt: it fits the model once on the first seven months and applies it frozen to the following four, although every new hour reaches the historian within the hour and refitting Extra Trees on the entire record takes about a second. Refitting at a fixed interval on all data logged so far, with every prediction still made strictly ahead of the data used to produce it, resolves the result. Out-of-time accuracy improves monotonically as the interval shortens, and at a daily refit R^2 on the withheld final 30% reaches 0.871 for TMP, 0.580 for water level and 0.574 for permeate flow, with TMP predicted to 0.008 bar, an 80% reduction

Table 3

Test set performance of 16 machine learning models for three target variables (TMP, permeate flow, membrane tank water level). Models ranked by mean R^2 across targets; best values in each column are underlined.

Model	TMP (bar)				Flow (m^3/min)				Level (%)			
	R^2	RMSE	MAE	ΔR^2	R^2	RMSE	MAE	ΔR^2	R^2	RMSE	MAE	ΔR^2
Extra Trees	0.988	0.010	0.005	0.012	0.933	0.066	0.052	0.067	0.908	0.341	0.227	0.092
Bagging	0.978	0.014	0.006	0.019	0.910	0.076	0.059	0.076	0.839	0.450	0.282	0.140
Random Forest	0.978	0.014	0.006	0.019	0.909	0.077	0.059	0.076	0.839	0.450	0.282	0.140
XGBoost	0.977	0.014	0.007	0.022	0.906	0.078	0.061	0.081	0.843	0.444	0.292	0.143
LightGBM	0.978	0.014	0.008	0.015	0.898	0.081	0.063	0.048	0.822	0.473	0.321	0.109
Hist Grad. Boost.	0.976	0.014	0.008	0.016	0.899	0.081	0.064	0.048	0.839	0.450	0.312	0.094
KNN	0.969	0.016	0.007	0.011	0.894	0.083	0.063	0.036	0.813	0.485	0.298	0.062
MLP Neural Net.	0.951	0.020	0.011	0.003	0.861	0.095	0.074	0.022	0.756	0.554	0.387	0.015
Decision Tree	0.942	0.022	0.010	0.042	0.785	0.118	0.086	0.096	0.650	0.663	0.383	0.240
GBoosting	0.937	0.023	0.014	0.011	0.824	0.107	0.084	0.024	0.698	0.616	0.437	0.059
SVR	0.893	0.030	0.016	−0.005	0.771	0.122	0.092	0.004	0.498	0.794	0.458	0.034
AdaBoost	0.781	0.042	0.037	0.002	0.655	0.150	0.120	0.023	0.308	0.932	0.796	−0.020
Lin Regression	0.499	0.064	0.045	0.015	0.469	0.185	0.147	0.010	0.195	1.006	0.696	0.001
Rid Regression	0.499	0.064	0.045	0.015	0.469	0.185	0.147	0.010	0.195	1.006	0.696	0.001
ElasticNet	0.480	0.065	0.046	0.005	0.435	0.192	0.151	0.008	0.159	1.028	0.735	−0.001
Las Regression	0.440	0.068	0.049	0.000	0.393	0.199	0.158	0.007	0.117	1.053	0.767	−0.002

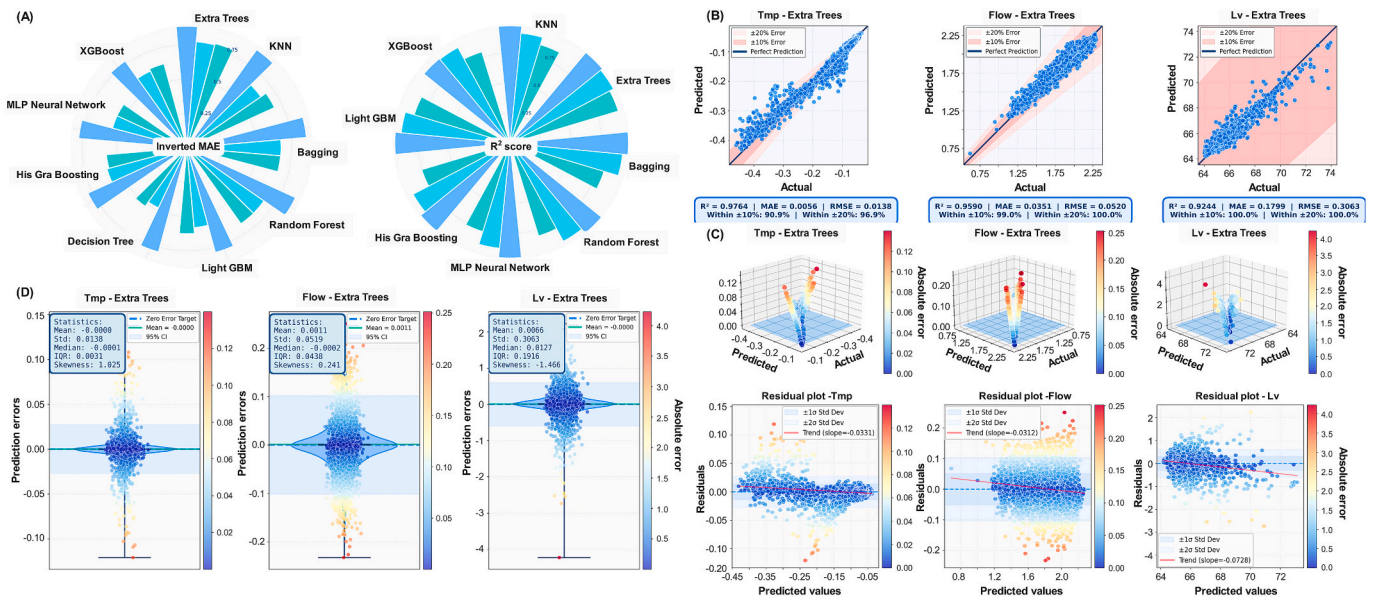


Fig. 5. (A) Model performance comparison, (B) Prediction quality, (C) Residual plot, (D) Error distribution analysis of the best-performing model (Extra Trees) on the independent validation dataset. Points on the 1:1 line and residuals scattered evenly about zero indicate that the model transfers to the independent stream without systematic bias.

against the frozen model (Fig. S17D, Table S12). Temporal generalisation is therefore governed by the refresh schedule rather than by any intrinsic limit of the model, and the deployment recommendation is concrete: refit on the accumulated record at least weekly, and daily where the historian connection makes it free.

3.2. Process interpretability analysis

3.2.1. Cross-model feature importance consensus and divergence

SHAP was applied across all 16 models and three targets, giving 48 feature-importance profiles (Figure 6A1–3; TreeSHAP–KernelSHAP comparison in Fig. S8A). The central result is a consensus one. Despite spanning six algorithmic families, every one of the 16 models ranked SRT first for TMP (consistency score 1.00) and 15 of 16 ranked it first for water level (mean rank 1.07 ± 0.25), the MLP being the sole exception; for permeate flow the picture is less unanimous, HRT ranking first in 7 of 16 models (2.28 ± 1.39) against 4.67 ± 1.89 for SRT. This convergence establishes that the dominance of SRT is a property of the data rather than of any single learner's inductive bias, a distinction the single-model analyses that constitute most of this literature cannot draw, and it survives the validation design, SRT remaining first-ranked for TMP and water level under blocked 24 h, blocked 168 h and chronological training even where predictive accuracy degrades. Flow rankings beyond the top two features are less stable and should be treated as indicative. Where the models diverge, the divergence is systematic and traceable to the explainer: linear models ranked C/N third or higher for TMP while every tree-based model ranked it last, and aeration ranked first for flow under KernelSHAP but fourth under TreeSHAP. KernelSHAP approximates Shapley values using random feature coalitions and so conflates correlated features [60], whereas TreeSHAP exploits the exact tree structure and is more reliable under correlation [61]; since Air and HRT co-vary here, higher aeration being deployed during higher-loading periods that also compress HRT, KernelSHAP absorbs part of the HRT contribution into aeration. All mechanistic discussion below therefore rests on the Extra Trees attributions.

3.2.2. Mechanistic interpretation of the dominant drivers

The dominance of SRT is the most physically interpretable finding of the analysis. SRT governs the composition, morphology and activity of the sludge community, and at this facility it ranged from 43.3 to 108.3

days, placing the entire operating range within the extended-aeration regime. Across the Extra Trees dependence and beeswarm plots (Fig. 6B1, Fig. S9), SHAP values are positive at 45–75 days, pass through a wide-scatter transition at 80–90 days where the effect becomes strongly conditional on co-occurring variables, and fall sharply beyond 90 days, reaching -0.3 to -2.0 at 100–108 days. That increasing SRT worsens TMP appears to contradict the expectation that longer sludge age reduces fouling through endogenous respiration and hydrolysis of EPS and SMP, but that expectation was established below 30–40 days. Here even the shortest SRT is already deep in the extended-aeration domain, so the EPS and SMP benefit is largely exhausted, and what changes between 45 and 108 days is accumulated biomass: less sludge is wasted, MLSS rises, and the accompanying increases in viscosity, cake thickness and cake compressibility outweigh any residual SMP reduction [61,62]. At very long SRT the floc becomes dominated by inert and recalcitrant particulates that deposit as a dense, poorly reversible layer, a mechanism with particular force in semiconductor wastewater where non-biodegradable fractions from photoresist and cleaning chemistries adsorb preferentially onto PVDF. The same logic explains the other targets: permeate flow is suppressed at the longest SRT as viscosity raises internal hydraulic resistance [63], and water level rises monotonically with SRT. The 80–90 day transition band is where operating choices decide the outcome, and the $\text{HRT} \times \text{SRT}$ interaction (0.056; Fig. S8B1) quantifies the coupling: at short HRT the biomass-driven penalty of rising SRT is partly offset, whereas at long HRT the two compound.

3.2.3. Hydraulic and aeration effects

HRT ranked second for TMP (2.33 ± 0.47) and first for permeate flow among the leading tree-based models, reflecting its dual role as a control on organic loading and on hydraulic capacity. Short HRT of 6–7 h yields positive SHAP for TMP and the contribution declines steadily above 7 h, reaching -1.3 to -2.2 at 10–11 h. The conventional expectation is the opposite, since short HRT should leave more residual SMP, but three features of full-scale operation reconcile it: short HRT corresponds to high throughput, which at this fab coincides with active production and a more dilute wastewater; long HRT periods above 9 h co-occur with reduced wasting and therefore with biomass accumulation; and during high-throughput operation the control system maintains more frequent backwash and relaxation cycles. For permeate flow the dependence is non-monotonic with an optimum near 6.5–7.5 h

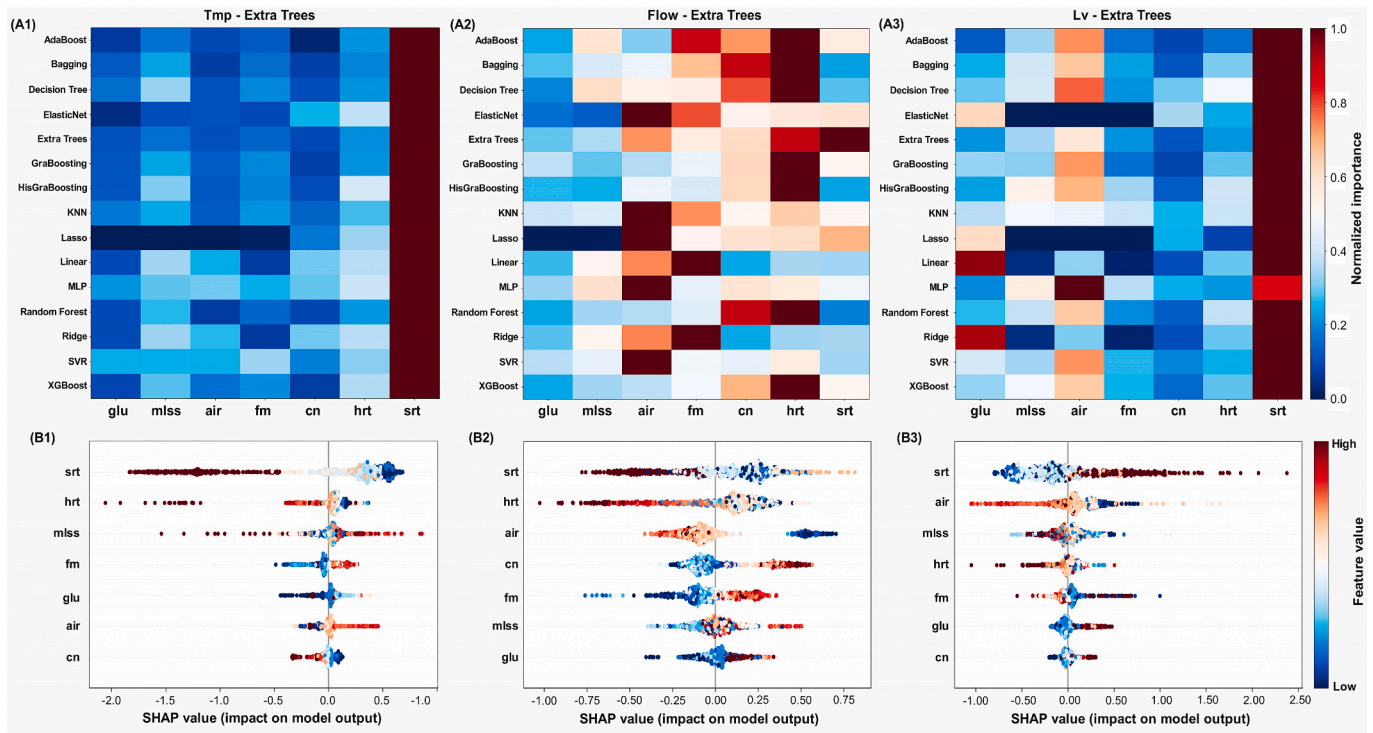


Fig. 6. (A) SHAP features importance comparison heatmaps and (B) Beeswarm plot across all 16 models for (1) TMP, (2) permeate flow, and (3) water level prediction. Cell values indicate mean |SHAP| for each model–feature pair. In the heatmaps each row is a model and each column a variable, so a column that is dark across every row marks a driver all 16 models agree on; in the beeswarm plots each point is one observation, its horizontal position the effect on the prediction and its colour the value of that variable.

before the hydraulic ceiling asserts itself, since $HRT = V/Q$ at fixed tank volume. The aeration rate ranked second for water level (0.216) and third for permeate flow but only sixth for TMP (0.063), an asymmetry following from its mechanism: it acts through hydrodynamic shear at the membrane surface, so it governs the hydraulic outputs more directly than TMP, which is additionally controlled by internal and adsorptive fouling that shear cannot dislodge. For TMP the dependence is non-monotonic, insufficient scouring below about 5000 m^3/h allowing fouling to proceed unchecked and the response bifurcating above about 6500 m^3/h , since high aeration relieves fouling only where the cake remains amenable to shear and is ineffective once long SRT and high MLSS have consolidated the layer [38]. For water level the large $Air \times SRT$ interaction (0.1318, the largest off-diagonal term across all targets; Fig. S8B3) shows that the benefit of additional aeration is amplified at low SRT and diminishes at high SRT [64].

The aeration rate also shows an apparent negative association with permeate flow. This is not a physical suppression of flux: in a pumped submerged MBR the flux is set by the suction pump, and the association instead reflects control logic, low aeration occurring during low-loading periods when the membrane is under minimal hydraulic stress while high aeration is deployed reactively during high loading or active fouling when flow is already constrained. TreeSHAP partially disentangles the confounding, ranking aeration fourth for flow behind HRT, SRT and C/N, but the residual association is mediated through covariation with HRT and SRT rather than any direct hydrodynamic effect. It is an instructive example of why dependence plots from observational plant data record association and not necessarily causal mechanism.

3.2.4. Secondary biological drivers

MLSS ranked third for TMP (3.33 ± 1.35) and for water level (3.87 ± 1.15), with an inverted-U dependence for TMP: near-zero contribution at 2000–3000 mg/L, a maximum near 6000–6500 mg/L, then a sharp

fall to -1.5 at 7000–7500 mg/L as cake resistance rises disproportionately, consistent with the near-quadratic scaling documented for PVDF hollow-fibre systems [65]; the $MLSS \times SRT$ interaction (0.042) confirms amplification at shorter SRT where EPS-rich sludge raises cake compressibility. F/M ranked fourth for TMP, with a maximum near 0.030 – 0.035 day^{-1} and a decline at the highest values where SMP overproduction accelerates gel-layer fouling, and a U-shape for water level with a steep rise above 0.05 day^{-1} . Glucose ranked fifth for TMP with a modest, largely indirect effect mediated through MLSS and F/M. C/N ranked seventh for TMP among the tree-based models, decreasing monotonically from positive contributions at 5–7 to -0.3 at 14–16, since carbon-rich, nitrogen-limited conditions favour carbohydrate-rich EPS and filamentous growth with poor filterability [66]; the divergence between model families for this variable is attributable to its correlation with MLSS and F/M and is a further illustration of the value of the cross-model comparison.

3.2.5. Interaction structure and methodological implications

Comparing TreeSHAP across the nine tree-based models with KernelSHAP across the seven non-tree models reveals consistent directional bias (Fig. S8A). For TMP the two agree on SRT dominance (0.530 against 0.458) and on HRT second (0.143), MLSS diverging most (0.144 against 0.109). For permeate flow the divergence is largest for aeration, first under KernelSHAP (0.243) and fourth under TreeSHAP (0.150) behind HRT (0.318), C/N (0.230) and F/M (0.160) [67]. For water level the two agree. The systematic pattern is that KernelSHAP overstates the leverage of readily adjustable short-term variables at the expense of the slower biological variables that dominate outcomes, reinforcing the use of TreeSHAP for operational guidance. The interaction matrices (Fig. S8) place the strongest couplings at $HRT \times SRT$ for TMP (0.056) and flow (0.0510), $MLSS \times SRT$ for TMP (0.042) and $Air \times SRT$ for water level (0.1318). These non-additive dependencies are the quantitative basis for treating the seven inputs jointly in Section 3.3 rather than optimising

them one at a time.

3.3. Optimal operating conditions for semiconductor MBR wastewater treatment

3.3.1. From feature importance to viable operating space

The interpretability analysis in Section 3.2 established, with consensus across 16 algorithms, which variables govern each performance indicator and in which direction they act. SRT is the dominant driver of TMP, worsening fouling monotonically across the facility's extended-aeration range through biomass accumulation, elevated viscosity and cake compaction; HRT sets the hydraulic ceiling on permeate flow; and aeration acts on TMP through a non-monotonic optimum with diminishing returns. What SHAP cannot supply is the set of input conditions under which all three targets are met at once, nor whether the direction that improves one target conflicts with the constraints of another. Section 3.3 addresses that gap, and it is the step that converts model insight into operating guidance.

Feasibility was mapped with the Extra Trees model using the manifold-constrained resampling scheme of Section 2.7, under three simultaneous constraints: TMP between -0.09 and -0.03 bar, permeate flow between 1.5 and 2.2 m³/min and water level between 65.0 and 67.0 % (Figs. S12–S13). Of 500,000 candidate states on the operating manifold, 207,238 (41.4%) satisfied all three. Ensemble dispersion across the 100 trees gives 95% prediction intervals of ± 0.008 bar for TMP, ± 0.17 m³/min for flow and ± 0.71 % for water level, so operation toward the interior of the envelope rather than at its edge is advisable for the two hydraulic targets in particular. Fig. 7 sets out the feasible range of each input against the attainment rate the plant actually achieved there. Two features of the scheme give the map its practical value. Every sampled state lies on the manifold the plant has explored, the median nearest-neighbour distance from sample to record being 0.121 standardised units against 0.119 between real hours, so every prediction is an interpolation among conditions the facility has genuinely operated and every recommendation is physically attainable. And windows are

graded by the conditional feasibility rate rather than by the number of feasible points, which identifies where the plant performs best rather than where it has spent the most time, and is what allows each recommendation in Table 4 to be stated with the frequency at which the plant has actually met its targets there.

The feasible region is not a rectangular box but a structured, correlated set. Fig. 8 maps the probability of meeting all three targets across every pair of operating variables, and in most pairs the high-probability region forms a narrow ridge rather than a rectangle, most clearly for MLSS against F/M, which run in strict opposition, and for C/N against HRT, where favourable operation follows a rising band. Treating the seven recommendations as independently adjustable set-points would therefore overstate the freedom available, and Fig. 8 is the reference for which combinations can be held simultaneously. In 20 of the 21 pairs the intersection of the two recommended bands lies in a high-probability region, with feasibility rates of 60–100% against a plant-wide rate of about 40%, so the per-variable recommendations compose without conflict. The single exception is a genuinely non-additive result that per-variable analysis cannot reach. Taken individually, SRT of 43.2–72.2 days and HRT of 5.875–6.381 h are the two strongest levers in the study, with measured attainment of 77.0% and 82.4%; taken together they are not. Only 35 logged hours combine both, and on those hours attainment falls to 34.3%, below the 37.1% plant-wide baseline. The reason is visible in the record: when SRT is held at 43.2–72.2 days the plant runs at an HRT of 6.2–7.5 h, whereas the 5.9–6.4 h band occurs at sludge ages of 72–108 days. Short HRT is a high-throughput condition, and sustaining high throughput at low sludge age demands a wasting rate the plant does not operate, so forcing both moves the system into a regime it has essentially never occupied. SRT and HRT should therefore not be set independently at their marginal optima: sludge age is the variable to fix first, being the slower of the two and the dominant driver of TMP, and HRT chosen conditionally on it, which is the structure of the joint rule validated in Section 3.3.4.

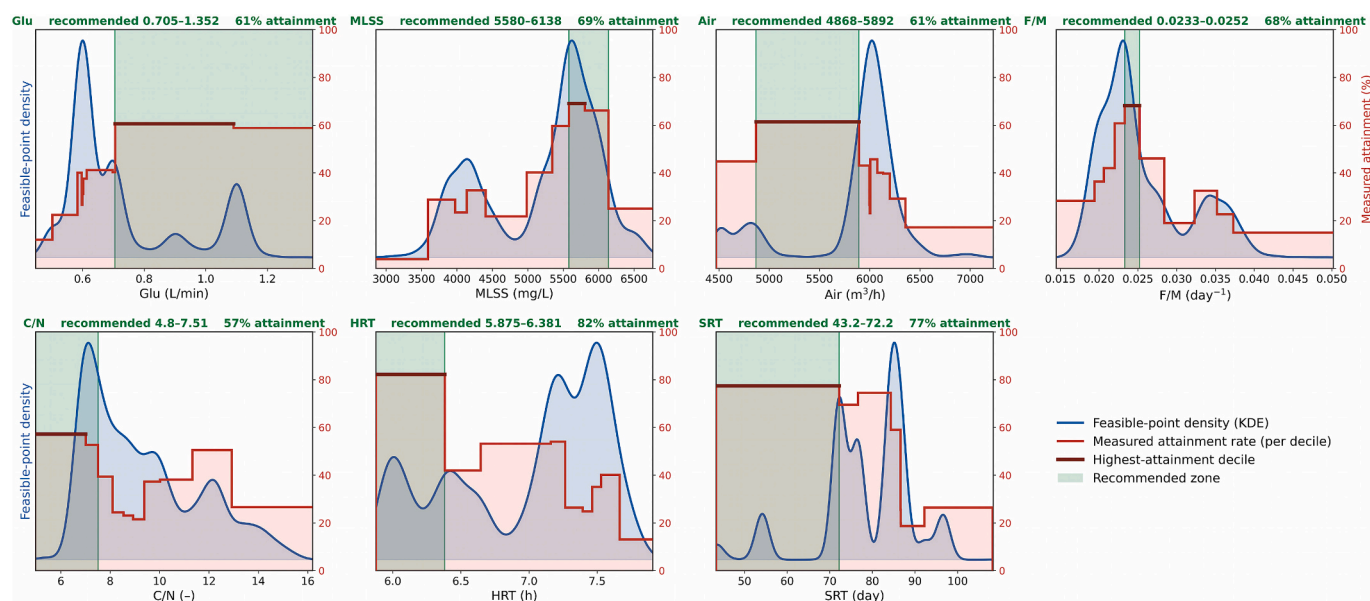


Fig. 7. Feasible region and measured plant performance for each of the seven inputs. Of the 500,000 sampled operating states, 207,238 (41.4%) satisfy all three performance targets simultaneously; the blue curve is a Gaussian kernel density estimate of their distribution, fitted to a random subsample of 15,000 of those states. The red step function is the measured attainment rate of the logged plant record, drawn across the true extent of each decile bin because the deciles differ widely in width, with the highest-attainment decile marked in dark red. The green band is the recommended operating zone of Table 4, which coincides with that decile by construction. Feasible-point density and attainment rate are distinct quantities and need not peak together: the density indicates where the plant has spent its operating time, whereas the rate indicates how often the three targets were met there. The recommended zone follows the rate. The blue curve shows where the plant can operate, the red step how often it met its targets there, and the green band is the recommendation, which follows the red step and not the blue curve. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 4

Recommended operating conditions for the target performance envelope (TMP -0.09 to -0.03 bar, permeate flow 1.5 – 2.2 m³/min, water level 65.0 – 67.0%). Zones are the contiguous bands maximising the conditional feasibility rate on the manifold-constrained sample. Measured attainment is the percentage of real logged hours inside and outside each band that met all three targets simultaneously; the plant-wide baseline is 37.1% . The bands are per-variable recommendations. They compose without conflict for 20 of the 21 variable pairs (Fig. 8); SRT and HRT interact, and the joint rule of Section 3.3.4 governs the choice of HRT once sludge age is fixed.

Parameter	Unit	Observed range	Recommended zone (rate-based)	Measured attainment inside / outside (%)	Operational note and out-of-time result
SRT	day	43.3–108.4	43.2–72.2	77.0 / 32.3	Hold below 85 d; target 43.2–72.2 d. Attainment falls monotonically to $<20\%$ above 87 d. Transferred out of time (+22.7 pp).
HRT	h	5.875–11.042	5.875–6.381	82.4 / 32.1	Operate 5.9–6.4 h. Attainment reaches zero above 7.95 h as the hydraulic ceiling caps permeate flow.
Air	m ³ /h	4470–7272	4868–5892	61.1 / 34.4	Keep within 4868–5892 m ³ /h; sufficient scouring for TMP without entering the reactive high-aeration regime.
MLSS	mg/L	1918–7630	5580–6138	67.6 / 29.5	Operate within 5580–6138 mg/L; the strongest out-of-time transfer of any lever (+30.9 pp).
C/N	–	4.80–17.52	4.80–7.51	55.0 / 32.6	Monitoring variable rather than a fixed set-point. Favourable band 4.80–7.51 in this record; because it is set by the fab discharge schedule it should be re-estimated periodically.
F/M	day ⁻¹	0.0117–0.0658	0.0233–0.0252	68.3 / 33.5	Keep below ~ 0.03 day ⁻¹ ; increase sludge extraction ahead of high-organic-loading sequences. Transferred out of time (+15.7 pp).
Glu	L/min	0.402–1.352	0.705–1.352	59.8 / 31.3	Conditional lever, dosed to maintain C/N; not an independent optimisation target.

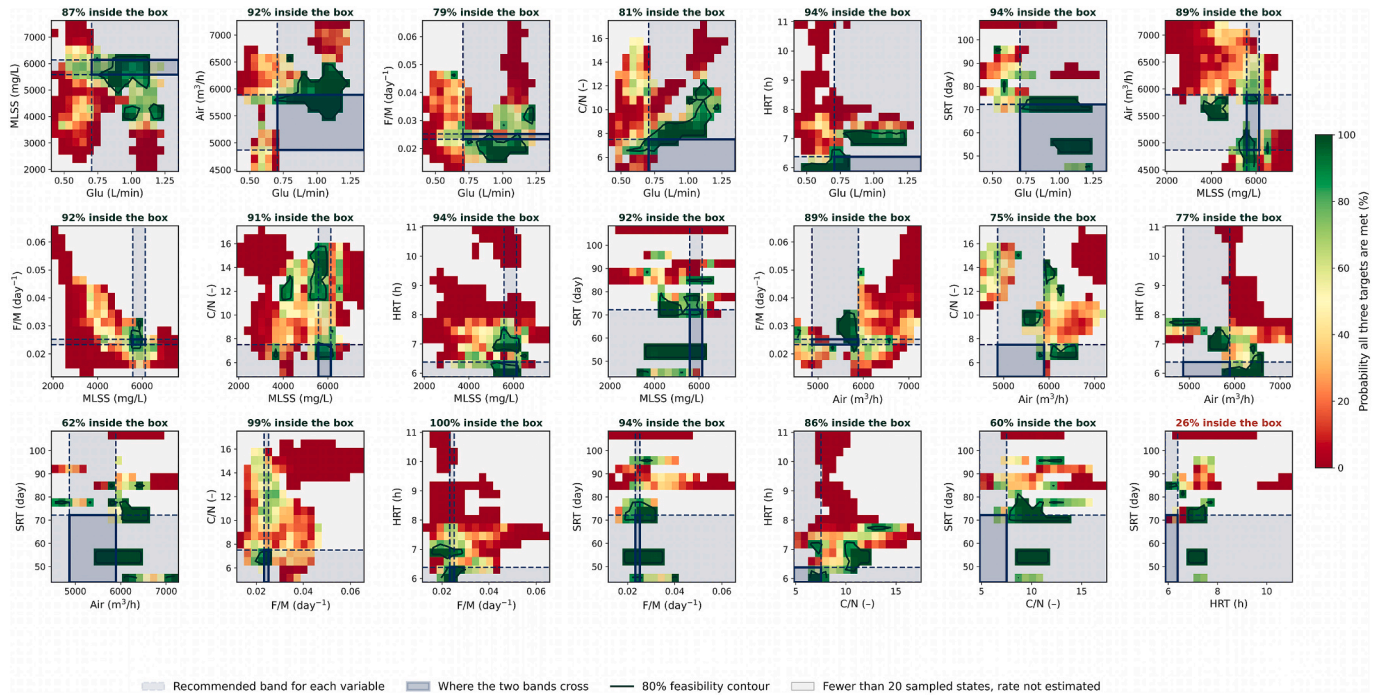


Fig. 8. Probability of meeting all three performance targets across every pair of operating variables, estimated from the manifold-constrained sample. Colour is the conditional feasibility rate $p(\text{all three targets met})$ within each bin, green indicating a high probability and red a low one; bins containing fewer than 20 sampled states are left grey. The dark green line is the 80% contour. The two navy dashed strips are the recommended bands for the variables on the two axes, taken from Table 4, and the outlined rectangle where they cross is the region satisfying both; the feasibility rate achieved inside it, computed over all sampled states it contains rather than bin by bin, is given above each panel. The crossing point of the two bands should sit on green. It does so in 20 of the 21 pairs; the single exception, HRT against SRT, identifies an interaction discussed in the text. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.3.2. Density structure of the feasible regions

The recommended operating window that follows is given in Table 4, where every row carries the measured attainment rate in the logged record alongside the model-derived rate, so that no recommendation rests on the model alone. HRT is the sharpest lever. Within the feasible region the conditional rate is 82.8% at 5.88 – 6.38 h and falls monotonically thereafter, reaching zero above 7.95 h; the measured attainment in the same bands is 82.4% and 0% (Fig. 7). This is the hydraulic ceiling identified in Section 3.2 expressed as an operating constraint: at fixed tank volume, HRT above roughly 8 h corresponds to influent flows at which the 1.5 m³/min permeate target cannot be met regardless of

membrane condition.

The aeration rate exhibits the opposing-target structure anticipated in Section 3.2. Measured attainment is highest at 4868 – 5892 m³/h (61.1%) and falls to 17.2% above 6356 m³/h (Fig. 9C). The TMP optimum sits at 5500 – 6000 m³/h, where sufficient shear keeps the cake mobile, while higher rates confer no further TMP benefit. The apparent decline of permeate flow with increasing aeration is, as established in Section 3.2, a consequence of control logic rather than hydrodynamic suppression of flux, so the recommended band should be read as the range in which the scouring requirement is met without the plant being driven into the reactive high-aeration regime that accompanies high

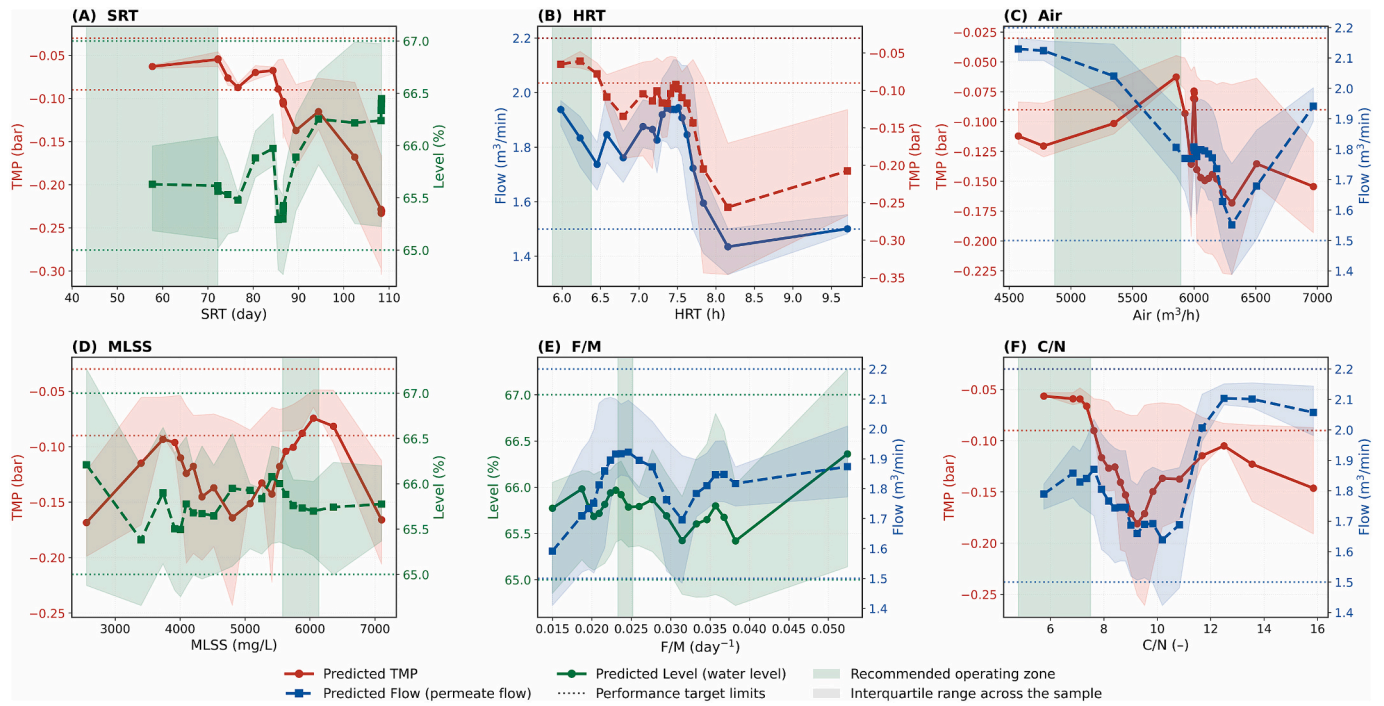


Fig. 9. Predicted output response across the manifold-constrained feasible region. (A) Effect of SRT on water level and TMP, (B) HRT on permeate flow and TMP, (C) opposing effects of aeration rate on flow and TMP, (D) non-monotonic effect of MLSS on TMP and water level, (E) F/M on water level and flow, (F) glucose dosing rate on water level and flow. Each panel shows how two predicted outputs respond as one input varies across its full range; the dotted horizontal lines are the performance limits, so a curve crossing them marks where that target fails, and the shaded band is the recommended operating zone.

loading. Correlations involving aeration warrant the same caution: the Pearson coefficient between Air and Flow is -0.410 over the raw record, and the stronger linear relationships in the dataset are MLSS-F/M

(-0.791) and SRT-TMP (-0.606).

The value of the framework rests on whether its feasibility surface corresponds to the plant's real behaviour, and the logged record contains

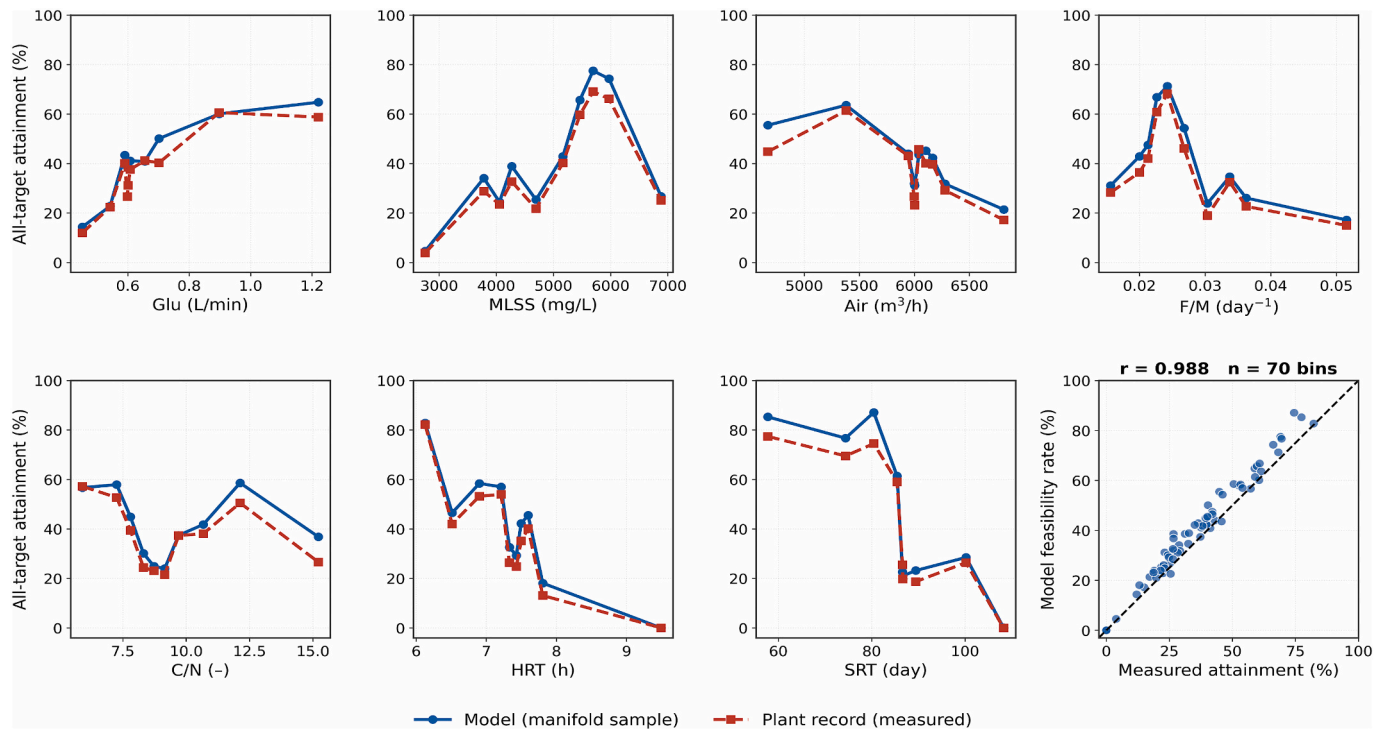


Fig. 10. Agreement between the model-derived conditional feasibility rate (blue) and the plant's measured target-attainment rate (red) across deciles of each of the seven inputs, with the paired comparison across all 70 operating bins shown in the final panel. The blue and red curves should lie on top of one another if the model reproduces plant behaviour, and in the final panel every point should fall on the dashed 1:1 line. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the ground truth needed to test this directly. Across the whole record, 1703 of 4593 h (37.1%) met all three targets simultaneously. Partitioning each input into deciles and comparing the model's predicted feasibility rate with the measured attainment rate in the same bin gives 70 paired comparisons, and the agreement is close: Pearson $r = 0.988$ ($p = 9 \times 10^{-57}$), Spearman $\rho = 0.984$, mean absolute deviation 4.3 percentage points and systematic optimism of only +4.2 points, with per-variable correlations between 0.96 and 0.998 (Fig. 10). The model therefore reproduces, quantitatively and without tuning to this comparison, how often the plant meets its targets as a function of where it is operating. This is the validation that licenses its use for operating guidance, and it is a more relevant test than forward prediction of a time series: the framework is asked which attainable operating states satisfy the constraints, and that is the question it answers accurately.

SRT shows the strongest and most monotone dose–response of the seven inputs. Measured attainment falls from 77.0% at 43.2–72.2 days, through 74.5% at 77–84 days and 59.0% at 84–86 days, to 18.7% at 87–92 days and 0% above 108 days; the model-derived rates track these values within a few percentage points across the whole range. The recommendation is therefore to hold SRT below approximately 85 days and to target the 43.2–72.2 day band, and it is supported by three independent lines of evidence pointing the same way: the cross-model SHAP consensus of Section 3.2 (Fig. 6), the model feasibility surface (Fig. 7) and the plant's own measured attainment (Fig. 11A), all of which place the optimum in the lower half of the operating range.

MLSS proves considerably more informative than a marginal reading of the feasible region suggests. Measured attainment rises from 3.9% below 3590 mg/L to a broad maximum of 67.6% across 5580–6138 mg/L, then falls to 25.1% above 6138 mg/L (Fig. 7, Fig. 9D, Fig. 10), and the model reproduces this inverted-U within 4 percentage points at every decile. The recommended band of 5580–6138 mg/L covers 918 logged hours, and it reflects the balance between maintaining sufficient biomass for treatment and avoiding the viscosity-driven cake resistance that dominates at higher concentrations. F/M behaves similarly, with a maximum of 68.3% attainment at 0.0233–0.0252 day⁻¹ against 15.0% above 0.0373 day⁻¹, consistent with SMP overproduction accelerating gel-layer fouling at high organic loading. The practical instruction is to keep F/M below approximately 0.03 day⁻¹ and to increase sludge extraction proactively when the production management system signals a high-organic-loading sequence.

C/N and glucose are best treated as conditional adjustments rather than primary set-points. Measured attainment for C/N is highest at

4.80–7.51 (55.0%) with a secondary shoulder at 11.3–12.9 (50.5%), a bimodal structure (Fig. 10) that reflects the two dominant discharge regimes at this facility: higher-carbon effluent during photoresist stripping and cleaning-agent discharge, and nitrogen-rich rinse water from polishing and etching steps. Glucose dosing is determined by the incoming C/N rather than optimised independently for membrane performance, and its association with tank level in the feasible region operates through the overall loading state rather than through F/M, with which it is very weakly correlated. The C/N association also proved the least stable of the seven over time, as reported in the out-of-time analysis below, and C/N guidance is therefore presented as a period-specific observation rather than as a transferable set-point.

3.3.3. Validation of the operating window on a withheld period

The most demanding test available for guidance derived from historical data is whether guidance obtained from one period improves outcomes in a later period that played no part in obtaining it. This involves no model prediction and is therefore independent of the accuracy assessment of Section 3.1. The record was divided chronologically, the window re-derived using only the first 70%, and its effect evaluated on the final 30% (Fig. 11B). The window from the first seven months places its principal condition on SRT (≤ 84.5 days) with HRT secondary, and identifies 415 h in the withheld period, of which 61.9% met all three targets against a baseline of 39.3% (odds ratio 3.89, Fisher exact $p = 3 \times 10^{-29}$). Lever by lever, the principal biological controls carry the result individually as well: MLSS transferred with a gain of 30.9 percentage points, SRT with 22.7 and F/M with 15.7 (Fig. S18). Sludge age, biomass concentration and organic loading, the three variables an operator can act on over hours to days, are therefore the levers for which the guidance is most firmly established, and the ones the protocol places at its centre.

3.3.4. Feature importance shifts and real-time control implications

Confining the SHAP analysis to the feasible region changes several importance rankings in ways that carry operational meaning (Tables S5–S7). These shifts are not artefacts: within the viable regime the most damaging conditions, namely SRT above 100 days, MLSS above 7500 mg/L and HRT above 10 h, are excluded by the output constraints, so shorter-timescale sources of variability become dominant. The clearest case is F/M, whose importance for water level rises by 87% from rank 5 to rank 4. In the full dataset, extreme SRT and high MLSS dominate level variance and mask the short-timescale contribution of organic loading; within the feasible region, where both are moderate, F/

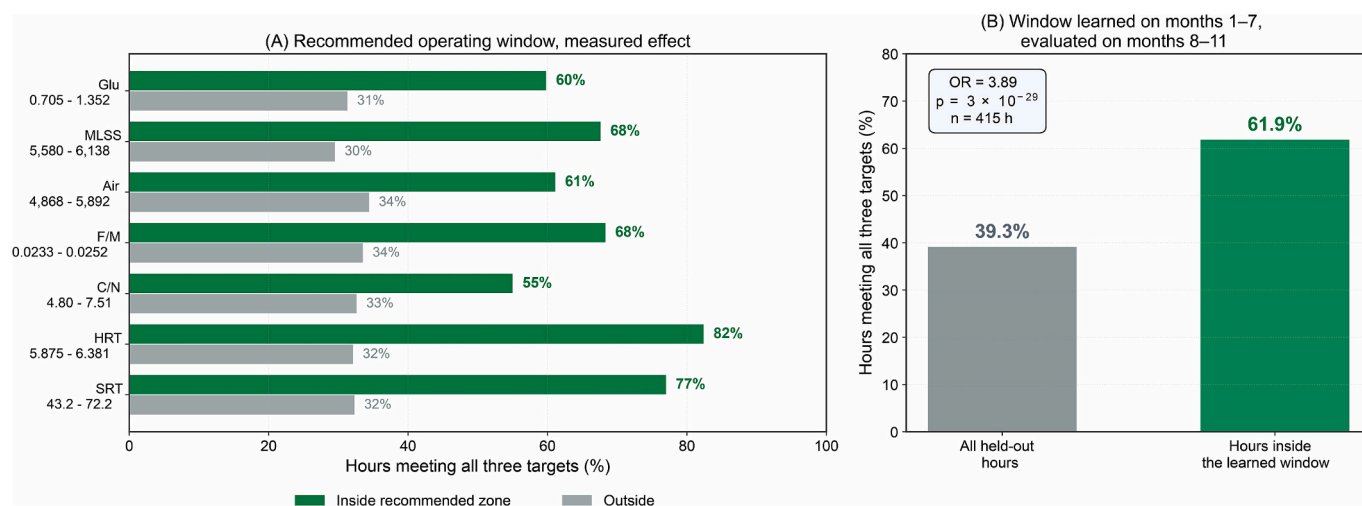


Fig. 11. Recommended operating window and its validation. (A) Percentage of real logged hours meeting all three performance targets inside and outside each recommended band. (B) Attainment across all held-out hours and inside the operating window learned on months 1–7 and evaluated on months 8–11. In panel A, a longer green bar than grey bar means the recommended band outperforms the rest of the record; in panel B the right-hand bar is the attainment achieved in months the analysis never saw. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

M emerges as the main driver of hour-to-hour level movement, following the fab's batch production schedule rather than any slowly adjustable biological parameter. With the parallel increase in MLSS importance for level, this identifies three fast-acting levers for level management: glucose dosing, responding in minutes, sludge extraction rate, in hours, and F/M adjustment through equalisation, also in hours (Fig. 9E). These allow operators to hold water level below the 67% threshold during high-loading production without waiting for the much longer SRT adjustment cycle. C/N also rises in rank for both TMP and flow; because its favourable band is set by the fab's discharge schedule and shifts as that schedule evolves, it is reported as a monitoring variable to be re-estimated periodically rather than a fixed set-point.

The recommendations in Table 4 do not rest on a single analytical method. Each is supported by the cross-model SHAP directional evidence of Section 3.2 across all 16 models, by the conditional feasibility surface of the manifold-constrained sample, and by the measured attainment rate in the plant's own record, with the primary levers additionally validated out of time. This convergence of model-based and measurement-based evidence, rather than the use of any one technique, is what distinguishes the present protocol from previous MBR optimisation studies, which typically report either importance rankings, or directional effects, or a feasible range, but do not reconcile all three against what the plant has actually achieved.

3.4. Comparison and perspectives

3.4.1. Predictive performance

Most prior studies focus on municipal or domestic wastewater treatment, often conducted at pilot or lab scales, and typically provide single-target predictions, such as TMP or fouling indicators (e.g., permeability, water flux, Table S8). For instance, [68] employed multilayer perceptron (MLP) and long short-term memory (LSTM) networks for TMP prediction in a large pilot-scale anaerobic MBR (AnMBR) treating municipal wastewater, achieving $R^2 > 0.91$. Similarly, [69] used random forest for fouling behaviour prediction in AnMBRs with $R^2 = 0.906$, while [70] applied adaptive neuro-fuzzy inference systems for water flux in osmotic MBRs, yielding R^2 between 0.9755 and 0.9861.

In contrast, the present study applies an Extra Trees regressor at full scale in a semiconductor wastewater plant, a domain of wide loading swings and stringent effluent requirements, and predicts three interdependent outputs simultaneously. Under a random partition, comparable to the designs used in the studies above, R^2 is 0.988 for TMP, 0.933 for permeate flow and 0.908 for water level; under a 24-h blocked partition that removes temporal near-neighbours, 0.830, 0.788 and 0.564. Both are reported because comparison with published single-target accuracies is meaningful only on a like-for-like partitioning basis, and the studies cited use random partitions of similarly autocorrelated records. Rather than claiming the highest accuracy reported to date, the appropriate conclusion is that a single ensemble learner achieves competitive accuracy on three coupled targets at full industrial scale without the computational overhead of recurrent architectures, and that the multi-target formulation addresses the operational reality that hydraulic dynamics and fouling cannot be optimised independently.

3.4.2. Optimization approach

The largest gap in the comparative literature lies not in predictive accuracy but in interpretability, which most comparison studies do not report at all. The framework used here combines kernel density estimation, feasible-region analysis, SHAP beeswarm and dependence analysis and Pearson correlation, applied in two analytical contexts: the full operational record (Section 3.2) and the feasible operating region (Section 3.3). This dual-context analysis is a methodological contribution in itself, because the importance ranking of several variables shifts when the analysis is confined to the viable regime, and those shifts have implications for real-time control that a single-context analysis cannot reveal. The 87% increase in F/M importance for water level within the

feasible region is the clearest example, identifying a near-real-time control lever that full-dataset SHAP leaves hidden behind the dominant slow variables.

Perhaps the sharpest distinction between the present study and the comparative literature lies in the meaning and scope of the term "optimization." In most comparative studies that report any optimization activity, the term refers to model optimization (Table S8) includes hyperparameter tuning architecture search, or training strategy selection. Only reference [71] applies optimization to operating conditions rather than model parameters, using a genetic algorithm to identify the input combination that minimizes a single fouling objective.

The present study is, to our knowledge, the first in the MBR machine-learning literature to perform multi-objective operating-condition optimisation within a manifold-constrained feasible input space, satisfying three membrane performance constraints simultaneously and mapping the joint structure of the 207,238 states that achieve this. This differs qualitatively from a single-objective genetic-algorithm search. Rather than returning one optimal point, the analysis produces a conditional probability surface over the attainable operating space, which allows the identification of regions where the system is most resilient to routine perturbation, distinct from the technically feasible boundary. The methodological point that emerges, and that we suggest is transferable well beyond this plant, is that a feasible region must be sampled on the joint operating manifold rather than across independent marginal ranges, and that operating recommendations must be graded by conditional attainment rate rather than by the density of feasible points.

The distinction between the feasible range and the high-attainment core is a practically significant contribution. For SRT the feasible envelope extends beyond 95 days while measured attainment above 87 days is below 20%; for HRT the feasible envelope reaches 7.65 h while attainment above 7.95 h is zero; for F/M the feasible envelope extends to 0.037 day^{-1} while attainment above 0.037 day^{-1} is 15%. An operator relying on the feasible boundary alone, that is on the output of a conventional constraint-satisfaction analysis, would conclude that these upper ranges are acceptable, when in practice they carry a high probability of violating one or more targets.

The rate-graded operating targets in Table 4, derived from 207,238 evaluated feasible states and cross-checked against 4593 logged hours, therefore provide a level of evidential support that is not available from a feasibility boundary alone.

3.4.3. Perspectives and future strategies

The structural contributions of this study, namely full-scale semiconductor wastewater application, simultaneous three-output prediction, benchmarking of 16 algorithms across six families under four validation designs, dual-context multi-model interpretability, and manifold-constrained optimisation validated against measured attainment and out of time, advance practice in three areas. Models trained on historical operating data from a complex industrial MBR can support operational guidance without additional laboratory characterisation, provided they are evaluated as response surfaces over the attainable operating space and validated against measured outcomes rather than on forecasting skill alone. Interpretability conducted within the feasible region rather than across the full historical envelope reveals control-relevant shifts in feature importance, a template applicable to any multi-output industrial process model. And the study delivers an evidence-graded operating protocol in which each recommendation carries an explicit measured attainment rate and an explicit statement of whether it transferred to a held-out period.

The boundaries of the framework should be stated alongside its contributions. The operating window has been verified three independent ways: against the plant's measured attainment across the operating space, against a three-month period withheld entirely from the analysis, and against an independent parallel membrane train. This is a stronger evidential base than a single-period, single-split analysis, but it remains a verification on historical data rather than a controlled trial, and part of

the association may be carried by variables outside the model, most obviously membrane age and cleaning history. A prospective trial alternating the recommended and incumbent strategies between the two parallel UF trains, so that influent conditions are shared, is the natural next step. Three further boundaries apply: the models were trained on one facility with a single PVDF UF configuration, so the method transfers but the numerical windows are site-specific; the framework characterises the manifold the plant has explored and is not intended to extrapolate to regimes absent from the record, with membrane age, cleaning frequency and effluent conductivity the most promising additional inputs; and SHAP attributions quantify association within the training distribution rather than physical causation, so confirming the proposed mechanisms would require membrane autopsy, zeta-potential measurement and filtrate organic fractionation.

4. Conclusion

This study set out to determine whether the operating record of a full-scale semiconductor-wastewater membrane bioreactor contains enough information to tell operators where to run the plant, and concludes that it does, provided the models are evaluated and used in the right way. Three findings support this. First, sludge retention time is the foundational determinant of membrane behaviour here, and the evidence is unusually robust: all 16 algorithms rank it first for transmembrane pressure, the ranking survives validation designs under which predictive accuracy itself degrades, and measured attainment falls monotonically from 77% at 43–72 days to below 20% beyond 87 days. Because the facility operates entirely within the extended-aeration range, the mechanism is biomass accumulation, elevated viscosity and cake compaction rather than the EPS and SMP dynamics of conventional short-sludge-age systems, so the expectation that longer sludge age mitigates fouling is inverted. Second, the question a plant record answers well is not what the plant will do next but which attainable operating states satisfy its performance constraints; mapped on the real joint operating manifold and graded by conditional attainment rate, the resulting surface reproduces measured attainment across 70 operating bins with a correlation of 0.99 and yields a window substantially narrower than the technically feasible envelope. Third, that window carries value beyond the period it came from: bands fixed on the first seven months raised simultaneous attainment from 39.3% to 61.9% over the following three, and refitted daily on the accumulated record the model predicts TMP to 0.008 bar with $R^2 = 0.871$ four months beyond its original training data, so temporal generalisation is governed by the refresh schedule rather than by any intrinsic limit of the model. The wider implication concerns how such models should be judged. Hourly industrial records are strongly autocorrelated, and accuracy measured on random partitions describes interpolation between neighbouring hours rather than generalisation to new operating conditions, whereas feature attributions prove stable across partitioning schemes. Interpretability, not headline accuracy, is therefore the property that survives scrutiny and should carry the operational weight, and the most useful output is not a predictor but a graded map of the operating space in which every recommendation carries the frequency with which the plant has actually achieved its targets there. The transition from reactive to proactive membrane management depends less on further gains in predictive accuracy than on validating guidance against outcomes the plant can verify and, ultimately, on prospective trials that test whether following such guidance causes the improvement these data associate with it.

CRediT authorship contribution statement

Yujae Jeon: Writing – original draft, Validation, Investigation, Formal analysis, Conceptualization. **Duc Anh Nguyen:** Methodology, Formal analysis. **Kim Anh Nguyen Thi:** Software, Investigation. **Quoc Thai Nong:** Software, Investigation. **Am Jang:** Writing – review &

editing, Validation, Supervision, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jwpe.2026.110865>.

Data availability

The analysis code, trained model configurations, SHAP outputs, feasibility-sampling routines and all derived result tables supporting this study are openly available at <https://github.com/nguyenducanh97/ML-MBR-semiconductor.git>. The repository includes complete benchmarking, temporal-validation, interpretability and manifold-feasibility pipelines together.

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